Experiment ‘p_3_2_classify_real’ Results

June 3, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` **:** 0
Responses containing `expect` **:** 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_3_2_classify_real
Natural language statement: Write a method that takes a floating point number and returns whether it is negative, zero, or positive. Add ”small” if the absolute value of the number is less than 1, or ”large” if it exceeds 1,000,000.
Method signature: p_3_2_classify_real (x: real) returns (s: string)

Ensures

- `x == 0.0 ==> s == "zero, small"`
- `1.0 > x > 0.0 ==> s == "positive, small"`
- `1.0 <= x < 1000000.0 ==> s == "positive"`
- `1000000.0 <= x ==> s == "positive, large"`
- `0.0 > x > -1.0 ==> s == "negative, small"`
- `-1.0 >= x > -1000000.0 ==> s == "negative"`
- `-1000000.0 >= x ==> s == "negative, large"`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method that takes a floating point number and returns whether it is negative,
    ↪ zero, or positive. Add "small" if the absolute value of the number is less than 1,
    ↪  or "large" if it exceeds 1,000,000.

The signature should be:

p_3_2_classify_real (x: real) returns (s: string)

The method should respect the following contract:

ensures x == 0.0 ==> s == "zero, small", ensures 1.0 > x > 0.0 ==> s == "positive, small
    ↪ ", ensures 1.0 <= x < 1000000.0 ==> s == "positive", ensures 1000000.0 <= x ==> s
    ↪ == "positive, large", ensures 0.0 > x > -1.0 ==> s == "negative, small", ensures
    ↪ -1.0 >= x > -1000000.0 ==> s == "negative", ensures -1000000.0 >= x ==> s == "
    ↪ negative, large"

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DmdjZD52VuE9RxsuUxeyFvix9x5ca
Model: gpt-5.5-2026-04-23
Created at: 1780484853
Finish reason: stop
Usage: 580 tokens in, and 593 tokens out

```
{
  if x == 0.0 {
    s := "zero, small";
  } else if x > 0.0 {
    if x < 1.0 {
      s := "positive, small";
    } else if x < 1000000.0 {
      s := "positive";
    } else {
      s := "positive, large";
    }
  } else {
    if x > -1.0 {
      s := "negative, small";
    } else if x > -1000000.0 {
```

```
      s := "negative";
    } else {
      s := "negative, large";
    }
  }
}
```

Final Program

```
method p_3_2_classify_real (x: real) returns (s: string)
        ensures x == 0.0 ==> s == "zero, small"
        ensures 1.0 > x > 0.0 ==> s == "positive, small"
        ensures 1.0 <= x < 1000000.0 ==> s == "positive"
        ensures 1000000.0 <= x ==> s == "positive, large"
        ensures 0.0 > x > -1.0 ==> s == "negative, small"
        ensures -1.0 >= x > -1000000.0 ==> s == "negative"
        ensures -1000000.0 >= x ==> s == "negative, large"
{
  if x == 0.0 {
    s := "zero, small";
  } else if x > 0.0 {
    if x < 1.0 {
      s := "positive, small";
    } else if x < 1000000.0 {
      s := "positive";
    } else {
      s := "positive, large";
    }
  } else {
    if x > -1.0 {
      s := "negative, small";
    } else if x > -1000000.0 {
      s := "negative";
    } else {
      s := "negative, large";
    }
  }
}
```

Total Token Usage

Input tokens: 580
Output tokens: 593
Reasoning tokens: 413
Sum of 'total tokens': 1173

Experiment Timings

Overall Experiment started at 1780484853556, ended at 1780484868258, lasting 14702ms (14.70 seconds)
Iteration #1 started at 1780484853557, ended at 1780484868257, lasting 14700ms (14.70 seconds)